

MANYBERRIES AGCM, AB, Canada

WMO: 710510

Lat: 49.3639N

Lon: 110.6781W

Elev: 927

StdP: 90.67

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-30.7	-26.6	-33.7	0.2	-30.6	-29.4	0.3	-26.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	32.2	16.2	30.3	15.9	28.2	15.7	18.2	27.0	17.3	26.6	16.5	25.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	12.2	20.6	14.2	11.3	19.9	13.1	10.5	19.0	55.2	26.8	52.2	26.6	49.6	25.9	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-36.7	35.5	3.4	1.6	-39.1	36.6	-41.1	37.6	-43.0	38.5	-45.4	39.7		
(5)			-36.8	19.9	3.3	1.3	-39.2	20.9	-41.2	21.6	-43.0	22.3	-45.5	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-9.3	-9.8	-2.1	4.5	10.3	14.6	18.6	18.0	12.8	4.8	-2.3	-9.5
	DBStd	11.87	7.69	8.40	7.59	5.04	4.33	3.67	3.33	3.80	4.91	5.35	7.08	7.78
	HDD10.0	2931	597	554	375	176	51	5	0	1	25	174	371	604
	HDD18.3	5234	856	787	633	417	250	123	37	54	178	418	620	862
	CDD10.0	849	0	0	1	9	60	142	267	248	108	14	1	0
	CDD18.3	110	0	0	0	0	1	11	46	43	10	0	0	0
	CDH23.3	2132	0	0	0	6	68	222	830	753	246	7	0	0
	CDH26.7	763	0	0	0	1	9	63	310	292	89	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	390	20	16	20	36	55	79	37	34	32	24	19	18
	PrecMax	591	39	39	37	66	95	173	146	88	63	59	44	39
	PrecMin	230	6	3	3	11	10	19	5	8	1	5	4	3
	PrecStd	72	8	9	10	15	22	37	29	21	17	13	9	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	9.8	17.3	23.4	27.9	32.4	34.2	34.2	32.3	24.2	16.9	9.1
		MCWB	3.5	5.3	8.3	11.0	14.3	16.8	16.9	16.4	15.2	13.3	8.6	4.4
	2%	DB	5.8	6.1	14.0	19.9	25.3	28.1	32.3	32.0	29.5	20.0	12.6	6.0
		MCWB	2.3	2.7	6.7	9.9	13.4	15.9	16.8	15.9	14.8	10.6	6.4	2.9
	5%	DB	4.0	3.8	11.0	16.9	22.7	25.7	30.2	30.2	26.0	17.0	9.6	3.8
		MCWB	1.5	1.3	5.2	8.0	12.1	15.0	16.4	15.6	13.9	9.4	5.0	1.4
	10%	DB	1.9	1.7	7.8	14.1	20.1	23.6	28.0	27.8	23.0	14.2	7.0	1.6
		MCWB	0.0	0.0	3.5	6.6	11.2	14.1	16.3	15.4	12.7	7.9	3.2	-0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	5.5	8.9	11.8	15.1	18.8	20.1	18.6	17.2	13.5	9.2	4.8
		MCDB	7.7	9.6	16.3	21.3	25.3	25.8	28.5	27.0	26.7	22.6	15.9	8.0
	2%	WB	2.6	3.1	7.2	10.2	13.9	17.2	18.6	17.5	15.7	11.6	7.1	2.8
		MCDB	5.5	5.6	13.0	19.5	23.7	26.7	27.3	27.3	26.2	18.0	11.8	5.5
	5%	WB	1.5	1.5	5.5	8.5	12.9	15.7	17.7	16.7	14.3	9.8	5.3	1.5
		MCDB	3.9	3.6	10.6	15.9	20.7	23.8	26.6	26.8	24.5	15.8	9.2	3.9
	10%	WB	0.0	0.0	3.7	6.8	11.7	14.7	16.9	15.9	13.1	8.3	3.5	-0.3
		MCDB	1.9	1.8	7.5	12.9	18.4	21.8	25.8	25.7	22.3	13.9	6.5	1.5
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.5	11.8	13.7	14.1	14.4	16.4	16.1	15.6	14.2	12.0	11.3
		MCDBR	13.0	13.5	16.0	18.6	18.9	17.8	19.2	19.8	20.7	18.5	15.6	13.9
		MCWBR	10.1	10.4	9.5	9.8	8.7	8.0	7.5	7.3	8.7	9.9	9.7	10.5
	5% WB	MCDBR	12.5	13.0	15.4	17.8	16.4	16.3	16.5	16.6	19.3	16.8	14.6	14.1
		MCWBR	9.9	10.0	9.5	9.4	7.9	7.8	7.4	6.5	8.5	9.3	9.3	10.7
Clear-Sky Solar Irradiance	taub	0.263	0.264	0.289	0.339	0.357	0.350	0.351	0.371	0.327	0.293	0.263	0.249	
	taud	2.421	2.479	2.476	2.338	2.333	2.406	2.418	2.332	2.433	2.517	2.504	2.436	
	Ebn at Noon	811	907	933	908	901	908	902	863	875	856	813	782	
	Edh at Noon	62	74	89	114	120	113	110	114	93	71	56	53	
All-Sky Solar Radiation	RadAvg	1.29	2.28	3.57	4.93	5.85	6.51	7.08	5.69	4.05	2.54	1.44	1.01	
	RadStd	0.11	0.18	0.26	0.33	0.44	0.38	0.36	0.36	0.32	0.24	0.09	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page